Geopolitical–Militaric Skirmish Learning Theory

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Abstract

This paper develops a unified theory of skirmish by synthesizing geopolitical influence, militaric scoring, and machine learning. A state is represented by a geopolitical score Z, a militaric residual score Y, and a skirmish gap $\Delta = |Z - Y|$. Skirmish is modeled as the observable manifestation of divergence between externally legible geopolitical influence and internally sustainable militaric capacity. The theory combines political economy, international relations, warfare economics, finance, statistics, and artificial intelligence. The paper derives consistency conditions for the skirmish equations, proposes supervised and Bayesian estimators, constructs graph-learning and reinforcement-learning extensions, and supplies algorithms for empirical implementation. The central claim is that skirmishes are not caused by power alone but by misalignment between geopolitical identity and militaric capacity.

The paper ends with "The End"

Keywords: skirmish, geopolitical influence, militaric score, machine learning, warfare economics, graph neural networks, political risk, international relations.

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1 Introduction

Skirmishes are small-scale, often unplanned military encounters. They usually occur below the threshold of declared war but above ordinary diplomatic tension. This paper develops a formal theory in which skirmish is explained by the gap between a state's geopolitical valuation and its militaric valuation.

The starting point is a geopolitical score

$$Z = \alpha \log M + \beta \log W + \gamma \log P,$$

where M is military size, W is wealth, P is population, and α, β, γ measure military, economic, and political weights. The militaric score is

$$Y = \alpha \log M + \beta \log(W - m) + \gamma \log(P - M),$$

where m denotes military wealth. The difference

$$\Delta = |Z - Y|$$

is interpreted as a structural misalignment between geopolitical projection and militaric sustainability.

The theory can be summarized by the equation

$$\text{Skirmish}_{AB,t} = F\left(\Delta_A, \Delta_B, o_A, o_B, \text{Var}(Z), \text{Var}(\phi), \theta_A, \theta_B, t\right),$$

where o_A, o_B are operational largeness coefficients, ϕ denotes angular geopolitical gaps, and θ_A, θ_B denote directional positions in a geopolitical circle.

2 The Geopolitical Circle

Let a state 0 have N neighboring states indexed by $i = 1, \ldots, N$. Each neighbor has a capital direction θ_i measured relative to a principal direction Ω.

Definition 1 (Geopolitical circle)**.** *The geopolitical circle of state* 0 *is the ordered angular system*

$$\mathcal{C}_0 = \{(\theta_i, M_i, W_i, P_i) : i = 1, \ldots, N\}.$$

The primitive geographic direction of influence is

$$\theta_{\text{geo}} = \frac{1}{N} \sum_{i=1}^{N} \theta_i.$$

Let the angular gaps be

$$\phi_i = \theta_{i+1} - \theta_i,$$

where angles are ordered cyclically and $\theta_{N+1} = \theta_1 + 2\pi$. The average angular gap is

$$\phi_{\text{avg}} = \frac{1}{N} \sum_{i=1}^{N} \phi_i.$$

The angular imbalance is

$$\text{Var}(\phi) = \frac{1}{N} \sum_{i=1}^{N} (\phi_i - \phi_{\text{avg}})^2.$$

Definition 2 (Geopolitical score)**.** *For neighboring state i, define*

$$Z_i = \alpha_i \log M_i + \beta_i \log W_i + \gamma_i \log P_i.$$

The regional geopolitical average is

$$Z_{\text{avg}} = \frac{1}{N} \sum_{i=1}^{N} Z_i,$$

and the regional geopolitical imbalance is

$$\text{Var}(Z) = \frac{1}{N} \sum_{i=1}^{N} (Z_i - Z_{\text{avg}})^2.$$

The Z-weighted direction of influence is

$$\Theta = \frac{\sum_{i=1}^{N} \theta_i Z_i}{\sum_{i=1}^{N} Z_i}.$$

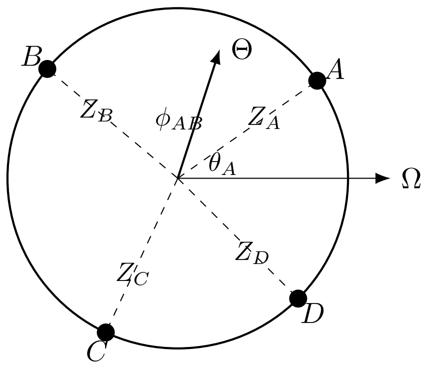

Figure 1: A geopolitical circle with neighbor directions, angular gaps, and Z-weighted direction of influence.

3 The Militaric Residual Score

The militaric score adjusts the geopolitical score by separating military resources from civilian resources.

Definition 3 (Militaric score)**.** *For state i, define*

$$Y_i = \alpha_i \log M_i + \beta_i \log(W_i - m_i) + \gamma_i \log(P_i - M_i),$$

where m_i is military wealth.

The domain restrictions are

$$M_i > 0, \qquad W_i > m_i > 0, \qquad P_i > M_i > 0.$$

Definition 4 (Skirmish gap)**.** *The skirmish gap of state i is*

$$\Delta_i = |Z_i - Y_i|.$$

Expanding,

$$Z_i - Y_i = \beta_i \left[\log W_i - \log(W_i - m_i)\right] + \gamma_i \left[\log P_i - \log(P_i - M_i)\right].$$

Therefore,

$$Z_i - Y_i = \beta_i \log\left(\frac{W_i}{W_i - m_i}\right) + \gamma_i \log\left(\frac{P_i}{P_i - M_i}\right).$$

Proposition 1 (Residual interpretation)**.** *If $\beta_i, \gamma_i > 0$, then $Z_i \geq Y_i$, and the skirmish gap satisfies*

$$\Delta_i = \beta_i \log\left(\frac{W_i}{W_i - m_i}\right) + \gamma_i \log\left(\frac{P_i}{P_i - M_i}\right).$$

Proof. Since $W_i > W_i - m_i > 0$,

$$\frac{W_i}{W_i - m_i} > 1,$$

and since $P_i > P_i - M_i > 0$,

$$\frac{P_i}{P_i - M_i} > 1.$$

Thus both logarithms are nonnegative. With $\beta_i, \gamma_i > 0$, the weighted sum is nonnegative. Hence $Z_i - Y_i \geq 0$, implying $\Delta_i = Z_i - Y_i$. □

4 The Skirmish Model

For two states A and B, define

$$S_A = o_A|Z_A - Y_A| = o_A\Delta_A,$$

$$S_B = o_B|Z_B - Y_B| = o_B\Delta_B,$$

where S_A, S_B denote skirmishers and o_A, o_B are operational largeness coefficients.

The original attenuation relation is

$$\frac{S_A}{S_B} = \frac{o_B}{o_A} e^{-At},$$

where $A > 0$ is the attenuation rate and $t \geq 0$ is time since the start of skirmish.

Proposition 2 (Consistency condition)**.** *If*

$$S_A = o_A \Delta_A, \qquad S_B = o_B \Delta_B,$$

and

$$\frac{S_A}{S_B} = \frac{o_B}{o_A} e^{-At},$$

then

$$\frac{\Delta_A}{\Delta_B} = \left(\frac{o_B}{o_A}\right)^2 e^{-At}.$$

Proof. From $S_A = o_A \Delta_A$ and $S_B = o_B \Delta_B$,

$$\frac{S_A}{S_B} = \frac{o_A \Delta_A}{o_B \Delta_B}.$$

Equating this with the attenuation equation gives

$$\frac{o_A \Delta_A}{o_B \Delta_B} = \frac{o_B}{o_A} e^{-At}.$$

Multiplying by o_B/o_A yields

$$\frac{\Delta_A}{\Delta_B} = \left(\frac{o_B}{o_A}\right)^2 e^{-At}.$$

□

A cleaner dynamic version is

$$S_A(t) = o_A \Delta_A e^{-a_A t},$$
$$S_B(t) = o_B \Delta_B e^{-a_B t},$$

so that

$$\frac{S_A(t)}{S_B(t)} = \frac{o_A \Delta_A}{o_B \Delta_B} e^{-(a_A - a_B)t}.$$

5 Political Economy Interpretation

The model has a political economy interpretation. The Z-score captures externally legible power: military size, wealth, population, and political stature. The Y-score captures militaric sustainability after subtracting military resource absorption from the civilian base.

Thus,

$$\Delta_i = |Z_i - Y_i|$$

measures the pressure produced when geopolitical projection outruns sustainable militaric foundations.

Table 1: Interpretation of core variables

Symbol	Meaning	Interpretation
Z_i	Geopolitical score	External influence and projection
Y_i	Militaric score	Internal militaric sustainability
Δ_i	Skirmish gap	Geopolitical–militaric misalignment
o_i	Operational largeness	Mobilization scale and logistical reach
θ_i	Direction angle	Spatial geopolitical orientation
ϕ_i	Angular gap	Geographic imbalance
$\mathrm{Var}(Z)$	Score variance	Regional power imbalance
$\mathrm{Var}(\phi)$	Gap variance	Spatial instability
A	Attenuation rate	Decay of skirmish intensity

6 Finance and Political Risk

In financial terms, Δ_i may be interpreted as a sovereign risk factor. A high Δ_i suggests that military commitments, border pressure, or geopolitical expectations exceed the state's internally sustainable base.

A reduced-form sovereign spread model may be written as

$$r_i - r_f = \lambda_0 + \lambda_1 \Delta_i + \lambda_2 \operatorname{Var}(Z) + \lambda_3 \operatorname{Var}(\phi) + \epsilon_i,$$

where r_i is the sovereign yield and r_f is the risk-free benchmark.

A currency-pressure model may be written as

$$\Delta e_i = \kappa_0 + \kappa_1 \Delta_i + \kappa_2 S_i + \kappa_3 C_i + u_i,$$

where Δe_i is exchange-rate depreciation, S_i is skirmish intensity, and C_i is capital-flow pressure.

7 International Relations Interpretation

The model gives four levels of conflict formation.

Table 2: Conflict stages in the skirmish theory

Stage	Condition	International relations interpretation
Latent imbalance	High $\operatorname{Var}(Z)$	Unequal regional power distribution
Directional tension	High $\operatorname{Var}(\phi)$	Spatially uneven neighborhood pressure
Militaric mismatch	High Δ_i	Projection exceeds sustainable militaric base
Skirmish	High $o_i\Delta_i$	Small-scale military encounter

The theory differs from simple balance-of-power theory because it does not claim that powerful states necessarily skirmish. Instead, it claims that skirmish probability rises when a state is mispriced by itself, its rival, or the regional system.

8 Statistical Learning Formulation

Let the dyad-year or dyad-month observation be

$$\mathcal{D}_{AB,t} = (X_{AB,t}, Y^{\text{obs}}_{AB,t}),$$

where $Y^{\text{obs}}_{AB,t} \in \{0,1\}$ indicates whether a skirmish occurred.

Define the feature vector

$$X_{AB,t} = \left[\Delta_A, \Delta_B, \frac{\Delta_A}{\Delta_B}, \frac{o_A}{o_B}, \operatorname{Var}(Z), \operatorname{Var}(\phi), |\theta_A - \theta_B|, t\right].$$

8.1 Logistic Skirmish Classifier

The probability of skirmish onset is

$$\Pr(Y^{\text{obs}}_{AB,t} = 1) = \sigma(w^\top X_{AB,t}),$$

where

$$\sigma(x) = \frac{1}{1+e^{-x}}.$$

The log-likelihood is

$$\ell(w) = \sum_{(A,B,t)} \left[Y^{\text{obs}}_{AB,t} \log \sigma(w^\top X_{AB,t}) + (1 - Y^{\text{obs}}_{AB,t}) \log(1 - \sigma(w^\top X_{AB,t})) \right].$$

8.2 Count Model for Skirmishers

Since skirmishers are counts, a negative binomial model is appropriate:

$$S_A \sim \mathrm{NegBin}(\mu_A, k),$$

with

$$\log \mu_A = c_0 + c_1 \log(o_A + \varepsilon) + c_2 \log(\Delta_A + \varepsilon) + c_3 \operatorname{Var}(Z) + c_4 \operatorname{Var}(\phi).$$

8.3 Survival Model for Duration

Let T be skirmish duration. A proportional hazard model is

$$h(t|X) = h_0(t) \exp(\eta^\top X).$$

A high attenuation rate means rapid decay:

$$S(t) = S(0)e^{-At}.$$

9 Machine Learning Extensions

The model can be estimated by several AI/ML methods.

Table 3: Machine learning methods for skirmish analysis

Method	Target	Use
Logistic regression	Skirmish onset	Interpretable probability model
Random forest	Skirmish classification	Nonlinear threshold discovery
Gradient boosting	Risk score	High predictive accuracy
Negative binomial regression	Skirmisher counts	Count prediction
Bayesian hierarchy	State heterogeneity	Uncertainty-aware estimation
Graph neural network	Regional contagion	Network-based skirmish prediction
Reinforcement learning	Crisis response	Policy simulation

9.1 Graph Neural Network

Let states be nodes in a graph

$$G = (V, E).$$

Each node i has features

$$h_i^{(0)} = [Z_i, Y_i, \Delta_i, o_i, \theta_i].$$

Each edge (i, j) has features

$$e_{ij} = [|\theta_i - \theta_j|, \phi_{ij}, \mathrm{trade}_{ij}, \mathrm{alliance}_{ij}].$$

A graph neural update is

$$h_i^{(k+1)} = \sigma \left(W_1 h_i^{(k)} + \sum_{j \in \mathcal{N}(i)} W_2 h_j^{(k)} + \sum_{j \in \mathcal{N}(i)} W_3 e_{ij} \right).$$

The predicted skirmish probability is

$$\widehat{p}_{ij} = \sigma \left(q^\top [h_i^{(K)}, h_j^{(K)}, e_{ij}] \right).$$

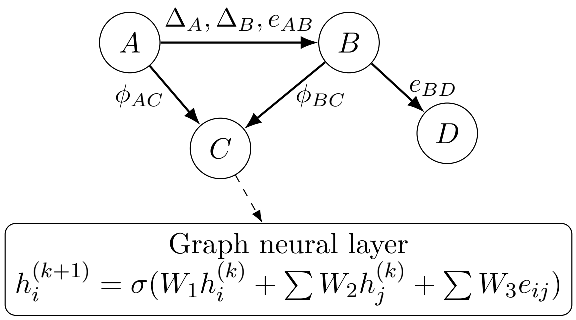

Figure 2: Graph-learning representation of regional skirmish propagation.

10 Bayesian Hierarchical Model

The coefficients $\alpha, \beta, \gamma, o, A$ may vary by state. A Bayesian specification is

$$\alpha_i \sim N(\mu_\alpha, \sigma_\alpha^2),$$
$$\beta_i \sim N(\mu_\beta, \sigma_\beta^2),$$
$$\gamma_i \sim N(\mu_\gamma, \sigma_\gamma^2),$$
$$o_i \sim \text{LogNormal}(\mu_o, \sigma_o^2),$$
$$A_i \sim \text{Gamma}(a_A, b_A).$$

The likelihood for skirmish onset is

$$Y_{ij,t}^{\text{obs}} \sim \text{Bernoulli}(p_{ij,t}),$$

where

$$p_{ij,t} = \sigma\left(w_0 + w_1\Delta_i + w_2\Delta_j + w_3 \operatorname{Var}(Z) + w_4 \operatorname{Var}(\phi)\right).$$

This allows the model to learn state-specific geopolitical psychology and operational behavior.

11 Reinforcement Learning Interpretation

A policymaker may choose actions

$$a_t \in \{\text{diplomacy}, \text{mobilization}, \text{de-escalation}, \text{alliance}, \text{sanction}\}.$$

The state vector is

$$s_t = [\Delta_A, \Delta_B, S_A, S_B, \operatorname{Var}(Z), \operatorname{Var}(\phi), r_A - r_f, r_B - r_f].$$

The reward function is

$$R(s_t, a_t) = -\rho_1 \Pr(\text{skirmish}) - \rho_2 E[S_A + S_B] - \rho_3 \text{FiscalCost}(a_t) + \rho_4 \text{DiplomaticGain}(a_t).$$

The optimal policy satisfies the Bellman equation

$$V^*(s) = \max_a \left[R(s, a) + \delta \sum_{s'} P(s'|s, a) V^*(s') \right].$$

Thus, reinforcement learning converts the skirmish model into a crisis-management engine.

12 Algorithms

Algorithm 1: Geopolitical–Militaric Feature Construction

Input: State data $(M_i, W_i, P_i, m_i, \theta_i)$, weights $(\alpha_i, \beta_i, \gamma_i)$, operational coefficient o_i.

1. Order neighboring states by angular direction θ_i.
2. Compute angular gaps $\phi_i = \theta_{i+1} - \theta_i$.
3. Compute $Z_i = \alpha_i \log M_i + \beta_i \log W_i + \gamma_i \log P_i$.
4. Compute $Y_i = \alpha_i \log M_i + \beta_i \log(W_i - m_i) + \gamma_i \log(P_i - M_i)$.
5. Compute $\Delta_i = |Z_i - Y_i|$.
6. Compute $\mathrm{Var}(Z)$, $\mathrm{Var}(\phi)$, and Θ.
7. For each dyad (i, j), construct $X_{ij,t}$.

Output: Machine-learning feature matrix X.

Algorithm 2: Skirmish Prediction

Input: Feature matrix X, skirmish labels Y^{obs}.

1. Split observations into training, validation, and test sets.
2. Estimate logistic model for skirmish onset.
3. Estimate negative binomial model for skirmisher counts.
4. Estimate survival model for skirmish duration.
5. Compare models using log-loss, AUC, calibration error, and predictive likelihood.
6. Select the model with best out-of-sample performance and acceptable interpretability.

Output: Predicted skirmish probability, expected skirmisher counts, and expected duration.

Algorithm 3: Reinforcement-Learning Crisis Policy

Input: State s_t, action set $\mathcal{A}$, transition model $P(s'|s, a)$, reward $R(s, a)$.

1. Initialize value function $V_0(s)$.
2. For each state s, evaluate all actions $a \in \mathcal{A}$.
3. Update
$$V_{k+1}(s) = \max_a \left[R(s, a) + \delta \sum_{s'} P(s'|s, a) V_k(s') \right].$$
4. Continue until $\|V_{k+1} - V_k\| < \epsilon$.
5. Choose
$$\pi^*(s) = \arg\max_a \left[R(s, a) + \delta \sum_{s'} P(s'|s, a) V^*(s') \right].$$

Output: Optimal crisis-management policy π^*.

13 Main Theorem

Theorem 1 (Skirmish as misalignment)**.** *Assume $o_i > 0$, $\beta_i, \gamma_i > 0$, $W_i > m_i > 0$, and $P_i > M_i > 0$. Then the expected skirmish intensity of state i is increasing in the civilian resource absorption ratios*
$$\frac{m_i}{W_i} \quad \textit{and} \quad \frac{M_i}{P_i}.$$

Proof. From the residual expression,

$$\Delta_i = \beta_i \log\left(\frac{W_i}{W_i - m_i}\right) + \gamma_i \log\left(\frac{P_i}{P_i - M_i}\right).$$

Rewrite as

$$\Delta_i = -\beta_i \log\left(1 - \frac{m_i}{W_i}\right) - \gamma_i \log\left(1 - \frac{M_i}{P_i}\right).$$

Let $x_i = m_i/W_i$ and $y_i = M_i/P_i$. Then

$$\Delta_i = -\beta_i \log(1 - x_i) - \gamma_i \log(1 - y_i).$$

Since

$$\frac{\partial \Delta_i}{\partial x_i} = \frac{\beta_i}{1 - x_i} > 0, \qquad \frac{\partial \Delta_i}{\partial y_i} = \frac{\gamma_i}{1 - y_i} > 0,$$

the skirmish gap increases in both military wealth absorption and military manpower absorption. Since

$$S_i = o_i \Delta_i$$

and $o_i > 0$, expected skirmish intensity also increases in both ratios. □

Corollary 1 (Skirmish pressure index)**.** *A parsimonious skirmish pressure index is*

$$\mathcal{K}_{ij,t} = \lambda_1 \Delta_i + \lambda_2 \Delta_j + \lambda_3 \operatorname{Var}(Z) + \lambda_4 \operatorname{Var}(\phi) + \lambda_5 |\theta_i - \theta_j|.$$

If all $\lambda_k > 0$, then skirmish pressure rises with militaric misalignment, geopolitical imbalance, angular imbalance, and directional rivalry.

14 Empirical Validation

The model can be validated using historical border incidents, militarized interstate disputes, defense expenditure, troop levels, GDP, population, bilateral trade, alliance data, and spatial information.

Table 4: Empirical validation design

Prediction task	Dependent variable	Evaluation metric
Skirmish onset	Binary incident indicator	AUC, log-loss, calibration
Skirmish size	Number of skirmishers	RMSE, negative log-likelihood
Duration	Days of skirmish	Concordance index
Escalation	Transition to larger conflict	Precision, recall, hazard ratio
Financial spillover	Yield or FX movement	R^2, MAE, directional accuracy

The strongest empirical test is whether Δ_i improves prediction beyond standard covariates such as GDP, military expenditure, population, alliance ties, regime type, and geographic contiguity.

15 Discussion

The theory has five major implications.

First, skirmish is not a simple function of military size. A state with a large military may be stable if its geopolitical and militaric scores are aligned.

Second, small states may skirmish if their geopolitical expectations exceed their residual militaric capacity.

Third, angular imbalance provides a spatial theory of instability. Conflict risk is not only dyadic but directional.

Fourth, financial markets may price skirmish risk through sovereign spreads, exchange rates, insurance premia, commodity routes, and defense-sector expectations.

Fifth, AI/ML methods are not merely predictive add-ons. They provide a way to estimate otherwise latent quantities such as operational largeness, attenuation, misperception, and regional contagion.

16 Conclusion

This paper has developed a unified geopolitical–militaric skirmish learning theory. The central object is the skirmish gap

$$\Delta_i = |Z_i - Y_i|,$$

which measures the divergence between geopolitical projection and militaric sustainability. The skirmish equation

$$S_i = o_i \Delta_i$$

then converts latent misalignment into observable skirmish intensity.

The theory integrates political science, economics, finance, international relations, warfare economics, statistics, data science, and AI/ML. Its key proposition is that skirmishes arise not from power alone but from mismatched power: the difference between the geopolitical role a state appears to occupy and the militaric base it can sustain.

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Glossary

Attenuation rate

The speed at which skirmish intensity decays over time.

Geopolitical circle
A circular representation of neighboring states by angular direction from a principal direction.

Geopolitical score
The score Z, measuring external geopolitical influence through military size, wealth, and population.

Militaric score
The score Y, measuring militaric capacity after subtracting military absorption from civilian resources.

Operational largeness
The coefficient o_i, measuring the scale at which a state can convert latent pressure into skirmishers.

Skirmish gap
The misalignment $\Delta_i = |Z_i - Y_i|$.

Skirmish pressure index
A composite index combining skirmish gaps, angular imbalance, geopolitical imbalance, and directional rivalry.

Angular gap
The directional separation ϕ_i between neighboring geopolitical positions.

Graph neural network
A machine-learning model that learns from states as nodes and geopolitical relations as edges.

Reinforcement learning
A learning framework in which a policymaker chooses actions to minimize skirmish risk and crisis cost.

The End